

ATI Perception Challenge – Tread Depth Estimation Design Plan Submission

Author: Amr Attia

Date: September 12, 2025

1. Problem Understanding

The goal is to design a system that estimates the tread depth of tires from a monocular video of a camera panning across the tire surface.

- **Input:** A video file (e.g., “Tire Pan 2.mp4”) showing a lateral sweep of the tire.
- **Output:**
 - Estimated tread depth in millimeters across the scanned region
 - CSV summary of depth over time

2. Approach Overview

I propose a hybrid pipeline combining interpretable groove geometry with optional monocular depth fusion (MiDaS). The system extracts groove-to-block distances in pixels and converts them to millimeters via calibration. MiDaS adds global depth context where grooves are faint or worn.

Pipeline Steps:

1. Frame Sampling
2. Preprocessing (CLAHE, blur)
3. Tire ROI Isolation
4. Groove Detection via Edge Projection
5. Depth Estimation (pixel spacing → mm)
6. Optional Fusion with MiDaS
7. Temporal Smoothing
8. Output CSV + Visual Overlays

3. Justification for Approach

- **Interpretable geometry:** Groove spacing is physically meaningful and explainable.
- **Modular fusion:** MiDaS adds robustness without sacrificing transparency.
- **Scalable sensing:** Works with just a camera; upgrades with calibration or structured light.
- **Real-time capable:** Lightweight enough for embedded deployment.

4. Data Considerations

- **Provided Data:** “Tire Pan 2.mp4”
- **Additional Needs:**

- Calibration image (e.g., checkerboard or ruler near tread)
- Optional structured light sweep (laser line)
- Ground-truth gauge readings for validation

5. Testing & Validation Strategy

Evaluation Metrics:

Metric	Description	Target
Mean Absolute Error	Avg. deviation from ground-truth depth	<0.5 mm
Repeatability	RMS variation across repeated scans	<0.2 mm
Confidence Score	Based on edge density + MiDaS agreement	≥ 0.8
Visual QC	Overlay inspection	Human-in-the-loop check

Validation Strategy:

- Manual gauge readings at known tread points
- Temporal consistency checks
- Visual overlays of groove columns and depth annotations
- Stress tests: blur, low light, worn tires

6. Edge Case Handling

Scenario	Strategy
Worn/faint grooves	Use MiDaS depth fusion
Wet/muddy tires	Cross-polarized lighting, MiDaS fallback
Perspective skew	Local homography rectification
Motion blur	Increase shutter speed, optional deblurring
Low contrast rubber	CLAHE + ridge filters

7. Sensor & Environmental Notes

- **Camera:** 1080p+, global shutter preferred
- **Lighting:** Grazing LED or cross-polarized ring
- **Motion:** Constant-speed lateral pan (~10–20 cm/s)
- **Calibration:** Checkerboard or ruler for pixel-to-mm conversion
- **Optional:** Line laser for structured light triangulation
 - Laser line projected at a fixed angle.
 - Stripe displacement → triangulated depth.
 - Sub-mm accuracy, robust for smooth/worn treads.

8. Implementation

Phase	Tasks
Prototype	Frame sampling, groove detection, calibration
Fusion	MiDaS integration, confidence scoring
QC & Export	Overlay generation, CSV output
Validation	Ground-truth comparison, stress testing

Risk Level: Low to medium. Most risk lies in calibration accuracy and groove visibility.

9. Future Extensions

- Structured light triangulation.
- Stereo depth fusion.
- Learned groove segmentation (DL).
- Temporal filtering (Kalman, Huber).
- Multi-camera fusion for full tire coverage.
- Uncertainty-aware depth maps.

10. Visual Pipeline Diagram

Video Frame → Preprocessing → ROI Mask → Groove Detection → Pixel Spacing → mm Conversion → (Optional MiDaS Fusion) → Temporal Smoothing → Output (Overlay + CSV)

11. Conclusion

This design combines **geometry-driven explainability** with **depth-aware robustness** to deliver accurate, repeatable tread depth estimates. The modular pipeline ensures:

- **Sub-mm precision with calibration.**
- **Field deployability with only a monocular camera.**
- **Upgrade path to structured light or stereo** for production-grade accuracy.

12. Extra Work Beyond Requirements

- MiDaS integration with drop-path patching.
- Metadata-based confidence scoring.
- CSV export with smoothed depth + confidence.
- Modular design ready for stereo/laser extensions.

13. Next Steps

- Capture calibration target for pixel-to-mm conversion.
- Run baseline groove-based pipeline and inspect overlays.
- Benchmark against manual gauge readings.
- Integrate MiDaS fusion and compare performance.
- Evaluate structured light feasibility for high-accuracy deployment.